ProFilm Meter

Professional Film Photography Light Meter & Director's Viewfinder

USER MANUAL · VERSION 4.14

Version 4.14 · Professional Film Photography Light Meter & Director's Viewfinder

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1. Getting Started

FIRST LAUNCH

On first launch, ProFilm Meter walks you through a five-step onboarding wizard covering the core controls, camera permission, and key settings. You can revisit this information at any time via **Settings** → **Help / Manual**.

CAMERA PERMISSION

The app requires camera access to meter light and display the viewfinder. When prompted, tap **Allow**. If you denied permission, go to your device's Settings app → Privacy → Camera → ProFilm Meter and re-enable it.

NAVIGATION

The app has **three tabs across the top** of the screen. Tap any tab label to switch:

- **Meter** — the main light meter and viewfinder screen
- **Journal** — your saved exposure log
- **Settings** — all configuration options

The active tab is highlighted in amber with an underline. There is no bottom navigation bar and no gear icon shortcut — Settings is always reached by tapping the Settings tab.

2. The Meter Screen

The Meter screen is your primary workspace. From top to bottom:

VIEWFINDER AREA

The live camera preview with overlaid frame lines, histogram, and status badges. Shows the scene as your film camera would frame it.

EV DISPLAY

The large centre readout shows the current **Exposure Value (EV)**. This updates continuously from the live camera feed.

- **Measured** — the raw metered EV before any compensation
- After locking, the display freezes and the calculated settings are shown

STATUS ROW

A horizontal row of coloured chips showing active features:

- **RECIP** (teal) — reciprocity failure is active and exceeds the threshold
- **FLASH** — flash peak is held
- **+Xmm** — bellows extra extension is active (number shows the mm entered)
- **ENV** — environmental compensation is active
- **SPECTRAL** — spectral sensitivity correction is being applied
- **SW** — software estimation is active (see Calibration section)
- **+/−** — manual exposure compensation is set

CONTROL BAR

A horizontal row of four tappable values sits between the EV display and the status row:

- **ISO/Film** — your film speed and active film stock
- **Aperture** — your f-stop (shown in amber when calculated by the meter in shutter priority)
- **Time** — your exposure time (shown in amber when calculated by the meter in aperture priority)
- **Comp** — manual exposure compensation (shown in amber when non-zero)

Tap any value to open its selector. For **Aperture**, **Time**, and **Comp**, a bottom sheet slides up with a vertical scroll-wheel dial. The selected value is highlighted in amber at the centre of the wheel.

The ISO/Film button opens a two-tab picker:

- **ISO tab** — scroll-wheel dial for choosing your film speed
- **Film Stock tab** — browse and select a film stock; your **Preferred Setups** appear at the top for one-tap recall, followed by the full stock list

Using the dial:

- **Scroll up or down** within the wheel to move through values — it snaps to the nearest stop
- **Tap the ↑ arrow** at the top of the wheel to step one stop up
- **Tap the ↓ arrow** at the bottom of the wheel to step one stop down
- **Tap any visible value directly** to jump to it immediately
- Each step triggers a light haptic tap

The step size for aperture and shutter is controlled by **Stop Increment** in Settings → Metering. When multiple increments are active, the wheel shows the merged set.

LOCK BUTTON

Tap **LOCK** to freeze the current reading. The live EV and calculated settings are captured at that moment. Tap **UNLOCK** to resume live metering.

When locked:

- Reciprocity correction is calculated and shown in teal above the shutter value
- A timer icon appears if the corrected time exceeds 2 seconds — tap it to open the Countdown Timer

SAVE BUTTON

Tap **SAVE** to log the current reading to the Field Journal. A confirmation screen lets you add notes and attach the viewfinder photo.

BOTTOM BUTTONS

Two rows of icon buttons run below the control bar and status chips.

Row 1:

- **Lock** — freeze / unfreeze the current reading
- **Grid** — cycle through viewfinder grid overlays (None → Rule of Thirds → Golden Ratio → Crosshair)
- **Flash** — arm/disarm flash metering (long-press to disarm)
- **Metering Mode** — cycle through Spot, Center-Weighted, and Average

- **Viewfinder / Format** (film icon) — open the lens and format picker sheet

Row 2:

- **Filters** — open the lens filter sheet
- **Bellows** — toggle bellows compensation on/off and enter extra extension beyond the focal length in mm
- **Orientation / Multi-Point** — toggle portrait/landscape, or add a point reading when multi-point is enabled
- **Save** — log the current reading to the Field Journal

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3. Metering Modes

Switch metering modes by tapping the **Metering Mode button** (row 1 of the bottom buttons). Each tap cycles to the next mode in the sequence: Spot → Center-Weighted → Average → Spot.

SPOT

Measures a small circle in the centre of the frame. Use for precise readings of a specific tone — a face, a shadow, a highlight. The spot area size is adjustable in **Settings → Metering → Spot Metering Size**.

> **Note:** *In spot mode, a* **Scan Scene** *button appears next to the Spectral badge — tap it to lock the spectral correction to your exact spot. For RA4 color analysis, spot metering is not recommended — use center-weighted or average for reliable color temperature readings.*

CENTER-WEIGHTED

Prioritises the centre of the frame while still considering the edges. Mimics the behavior of traditional in-camera meters. Area size is adjustable in **Settings → Metering → Center-Weighted Size**.

AVERAGE

Reads the entire frame evenly. Best for scenes with consistent, even lighting. The area size controls how much of the centre is emphasised vs. the full frame average (**Settings → Metering → Average Metering Size**).

4. Priority Modes

Priority mode is set automatically by which value you tap in the Control Bar:

ACTION	MODE SET	WHAT IS CALCULATED
Tap **Aperture** in the control bar	Aperture Priority	Shutter speed (shown in amber)
Tap **Time** in the control bar	Shutter Priority	Aperture (shown in amber)

The calculated value updates live as you change the fixed value. The amber colour on the computed value is a reminder that the meter chose it — you did not set it manually.

5. Multi-Point Metering

Capture multiple readings and average them into a single exposure value — useful for scenes with mixed lighting where you want to balance highlights and shadows.

To enable: Settings → Metering → Multi-Point Metering. Choose how many points to collect (2–4).

Workflow:

1. Enable multi-point and set the count.
2. On the meter screen, point at your first subject and tap to record that EV.
3. Move to each subsequent area and tap again.
4. Once all points are collected, the meter locks automatically and shows the averaged EV.

In the Field Journal, tap the multi-point chip on any saved entry to expand an inline panel showing the individual EV recorded at each point (P1, P2, P3 …).

6. Viewfinder

The viewfinder simulates your film camera's field of view based on your chosen film format and focal length.

FILM FORMAT

Tap the **film icon button** (Viewfinder/Format) in the bottom bar row 1. A sheet opens with two options — tap **Film Format**. A vertical scroll-wheel dial lists all your enabled formats. Scroll to the one you want and it is selected immediately. The viewfinder frame lines scale to match the aspect ratio. Manage available formats in **Settings** → **Gear Management** → **Formats**.

FOCAL LENGTH

Tap the **film icon button** (Viewfinder/Format) in the bottom bar row 1 and choose **Focal Length**. A vertical scroll-wheel dial lists all your enabled focal lengths. Scroll to your taking lens. The viewfinder magnification adjusts so the framing matches what that lens sees on film.

CAMERA LENS

On iOS, switch between the phone's physical camera lenses (ultra-wide, main, telephoto) in **Settings** → **Viewfinder** → **Camera Lens**. On Android, the standard wide lens is used.

GRID OVERLAYS

Tap the **Grid button** in bottom bar row 1 to cycle through:

- **None** — clean viewfinder
- **Rule of Thirds** — 3×3 grid
- **Golden Ratio** — Phi-based grid
- **Crosshair** — centre cross

TAP TO FOCUS

Tap anywhere in the viewfinder to set the autofocus point. A focus ring animation confirms the tap.

LOCK FREEZE

When you tap **LOCK**, the live viewfinder preview freezes to a still image captured at that exact moment. This lets you study the framing or composition that you metered. The live preview resumes when you tap **UNLOCK**.

VIEWFINDER MARGIN

Adjust the inner margin of the frame lines via **Settings** → **Viewfinder** → **Viewfinder Margin** (0–20%). Useful to simulate camera-specific viewfinder coverage.

ORIENTATION

Set the sensor orientation to **Portrait** or **Landscape** in **Settings** → **Viewfinder** → **Orientation**.

7. Viewfinder Zoom Slider

The vertical amber slider on the **left edge of the viewfinder** lets you change your film lens focal length on the fly to scout compositions without touching the focal length dial.

HOW TO USE

- **Drag up** — increases focal length (zooms in, narrower field of view)
- **Drag down** — decreases focal length (zooms out, wider field of view)
- Both the viewfinder framing and the metering update live while dragging
- The focal length label next to the handle shows the current value as you drag

SCALE & RANGE

- The scale is **logarithmic** spanning **25mm to 1500mm** — every 80 pixels of drag shifts focal length by one stop (×2 or ÷2)
- The small pip at the centre of the track marks approximately **194mm** on the absolute scale

ENABLING / DISABLING

- **Quick toggle:** Settings → Viewfinder → **Zoom Slider** — turns the slider on or off for the current session
- **Hide entirely:** Settings → App Tools Configuration → **Viewfinder Zoom Slider** — removes both the slider and its Settings row

8. Auto Lens Selection

When enabled, ProFilm Meter automatically switches to the best phone camera lens as you change your film lens focal length, choosing whichever lens requires the least digital zoom to match your framing.

Enable: Settings → Viewfinder → **Auto-select camera lens**

HOW THE SELECTION WORKS

The app compares your film focal length against the native equivalent focal lengths of your phone's available lenses (ultra-wide ≈ 13 mm, main ≈ 26 mm, telephoto ≈ 77 mm). It picks the lens with the largest native focal length that is still shorter than or equal to your film focal length — so the camera always zooms in slightly rather than out, preserving image quality.

FILM FOCAL LENGTH	LENS SELECTED
20 mm	Ultra-wide (13 mm)
35 mm	Main (26 mm)
85 mm	Telephoto (77 mm)

FLASH MODE

When you arm flash metering, the app automatically switches to the main (1×) lens regardless of your current focal length. Telephoto lenses have a narrower angle of view and can miss off-axis flash pulses. The previous lens is restored when you disarm.

CALIBRATION REQUIREMENT

Auto lens selection only switches to lenses that have been calibrated. If the ideal lens is uncalibrated, a nudge banner prompts you to calibrate. Tap **Calibrate** on the banner to open the calibration guide for that lens.

9. Histogram

The RGB histogram overlay in the viewfinder shows the distribution of red, green, and blue channel values across the current frame.

- **Clustered left** — underexposure
- **Clustered right** — overexposure
- **Spread across** — full tonal range

The histogram updates continuously from the live camera preview and is automatically hidden during flash mode to prioritise capture speed.

10. Scene Color Temperature

When enabled, a live Kelvin (K) badge appears on the viewfinder showing the estimated scene color temperature derived from the RGB balance of the live camera frame.

RANGE	LIGHT SOURCE
2700–3200 K	Incandescent / tungsten
5500–6500 K	Daylight
7000 K+	Open shade / overcast

When RA4 Color Reversal is enabled, this value also drives the CMY filter pack recommendations.

Enable: Settings → Preferences → **Show Color Temperature**

11. Flash Metering

ARMING

Tap the **Flash button** in bottom bar row 1 to arm flash detection. The button label changes to **Hold to Disarm** when armed.

HOW IT WORKS

Once armed, the current ambient EV is captured and locked as a reference. Fire your flash unit; the meter detects the peak brightness spike and calculates the correct exposure for that burst. A **FLASH** chip appears in the viewfinder and the EV display holds the flash reading. The ambient EV is shown below for comparison.

If **RA4 Color Reversal** is enabled, a separate color temperature analysis runs on the flash peak frame and appears with an amber-bordered badge.

RE-ARMING

Tap **Hold to Disarm** (or the flash icon again) to re-arm for the next shot, resetting the peak hold.

FLASH TRIGGER

If a Flash Trigger mode is configured (see Section 12), a **Trigger** button appears in the viewfinder. Tap it to fire the flash signal — the meter then automatically records the peak.

FLASH SENSITIVITY

Access: Settings → Metering → **Flash Sensitivity**

Flash Sensitivity controls how large a brightness spike the meter must see before classifying a frame as a flash event. A lower threshold catches weaker flashes; a higher threshold avoids false triggers from bright ambient light.

> **Important:** *After changing sensitivity,* **re-arm the flash** *(disarm then re-arm) for the new setting to take effect.*

LEVEL	SPIKE RATIO	WHEN TO USE
Maximum	1.05	Catches a 5% brightness increase — even a faint flash bounce through diffusion. Use for heavily diffused strobes, very weak fill flash bounced off a far ceiling, or low-power pops. Risk: may false-trigger on bright window glare, reflections, or moving headlights.
High	1.08	For subtle fill flash, ratio lighting (main + low-power fill), or when bouncing through thin diffusion panels.
Normal *(default)*	1.10	Balanced for most controlled studio and location flash work. A 10% spike is well above typical ambient variation in a studio or shaded outdoor location.
Reduced	1.15	Use outdoors in bright conditions (daylight fill flash), near bright windows, or when you are getting occasional false triggers at Normal.
Low	1.25	Use in highly variable ambient light — strong sunlight with cloud gaps, shooting near moving water or reflective surfaces. Only a clear, direct flash hit registers.
Minimum	1.50	Use only when you keep getting false triggers at all other settings — bright sun reflecting off glass, moving vehicle headlights, or high-contrast flicker lighting. Requires a strong direct flash hit to register.

Practical tips:

- Start at **Normal** and drop to **High** or **Maximum** only if the flash isn't registering
- Move up to **Reduced** or **Low** if you get false triggers from ambient light changes
- In a controlled studio with no ambient light variation, **Normal** or **High** always works
- Outdoors in full sun, start at **Reduced**; if still false-triggering, try **Low**

12. Flash Trigger Modes

Access: Settings → Metering → **Flash Trigger**

When a trigger mode other than **Off** or **Light Sensor** is selected and flash is armed, a **Trigger** button appears in the centre of the viewfinder. Tap it to send the fire signal — the meter then automatically records the peak.

OFF

No trigger signal is sent. Fire your flash manually at any time while the meter is armed. The meter will detect the peak spike whenever it occurs.

When to use: Any situation where you control the flash shutter sync yourself — a standard sync cable plugged into the camera, a flash connected to the camera hot shoe, or a radio trigger fired by the camera. The app just meters the flash, it doesn't need to fire it.

Hardware needed: None.

OPTICAL SLAVE (LED)

The phone's rear LED torch fires a short burst of light to trip the optical slave sensor built into most studio strobes and many speedlights. No cable needed.

When to use: When your strobe has an optical slave (S1 or S2) mode and you can't or don't want to run a sync cable. Great for portable location setups, older studio strobes without radio triggers, and any time you want a completely wireless test.

Hardware needed: A flash unit with an optical slave (S1/S2) mode. Nothing else required.

Torch Duration: The length of the LED burst. Configurable in Settings → Metering → **Torch Duration**:

DURATION	USE
50 ms	Fast slaves, short-duration strobes
100 ms	Most standard optical slaves
150 ms *(default)*	Reliable for most units, including older slaves
200 ms	Sluggish or heavily diffused slaves, or those needing a longer trigger pulse

Tips for best results:

- Point the phone toward the flash unit's optical sensor window — within 45° of axis
- Keep the phone within ~3–5 metres of the sensor for reliable triggering in bright ambient light
- In very bright sunlight, optical slaves may not respond — consider Cable Sync or radio triggers instead
- S2 mode on most strobes ignores the pre-flash from digital cameras; S1 mode fires on any light pulse

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CABLE SYNC (3.5MM)

The app plays a sharp audio click through the headphone jack to simulate a PC sync pulse. This voltage change on the audio channel can trigger compatible flash units via a sync cable adapter.

When to use: When you have a flash unit with a standard PC sync port and a USB-C → 3.5mm adapter (or 3.5mm headphone jack on older phones). Useful for older studio packs and monolights that predate radio triggers.

Hardware needed:

- USB-C → 3.5mm audio adapter (the standard inexpensive dongle)
- A 3.5mm → PC sync adapter cable (available from camera shops and online for a few dollars)
- A flash unit with a PC sync (male coaxial sync) input

Note: This is an experimental feature. The audio signal is not a true electrical sync pulse — it is an audio transient. Compatibility depends on the impedance and trigger threshold of the specific flash unit. Test on low power first.

LIGHT SENSOR *(ANDROID ONLY)*

Uses the device's hardware ambient light sensor to detect the flash spike automatically. The sensor responds in milliseconds — faster than the camera — and confirms the flash event for more reliable capture.

When to use: The simplest and most reliable Android option when your flash is close enough to the phone that the room brightens noticeably. No trigger button, no cable, no optical pointing — just arm, fire, and the sensor catches it.

Hardware needed: None. Not available on iOS (iPhones do not expose the ambient light sensor to apps).

Tips: Position the phone with the ambient light sensor (usually near the earpiece or on the front face) visible to the flash or at least in the same room. Works best with bare strobes or large softboxes; less reliable through heavy diffusion or bounce off distant surfaces.

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13. Film Stocks & Reciprocity Failure

SELECTING A FILM STOCK

Open the film stock picker from the **ISO/Film button** on the Control Bar — tap it, then switch to the **Film Stock tab**. You can also reach it via Settings → Gear Management → Film Stocks. Tap any stock to select it.

Your **Preferred Setups** appear at the top of the Film Stock tab for one-tap recall. The full stock list follows beneath them.

Selecting a stock applies its native ISO and links its reciprocity profile automatically. If you manually change the ISO dial after selecting a stock, the stock is deselected and your previous settings are restored.

ZEBRA DRY PLATES COLLECTION

The app includes the full Zebra Dry Plates catalogue (ISO 2, blue-sensitive). These include dry plates, tintypes, and golden tintypes. Zebra stocks show a **cart icon** linking directly to the Zebra store to purchase.

Tap **Zebra Dry Plates Film Stocks** in the stock picker to browse and add stocks from the live catalogue (fetched from GitHub, cached 24 hours).

Selecting a tintype stock automatically enables **Positive Exposure** (+0.5 stop).

RECIPROCITY FAILURE

Film loses sensitivity at long exposure times — a phenomenon called reciprocity failure. Each emulsion has a Schwarzschild exponent (p-factor) describing how severe this is.

Workflow:

1. Select a reciprocity profile in **Settings → Gear Management → Film Stocks → Reciprocity Profile**
2. During live metering the shutter always shows the raw calculated time. When it crosses the profile's threshold, a teal **RECIP** chip appears
3. Tap **LOCK** — the corrected duration (calculated from the locked time using the film's p-factor) appears in teal above the raw shutter value
4. Set your camera to the corrected time
5. If the corrected time exceeds 2 seconds, a timer icon appears — tap it to open the Countdown Timer pre-loaded with the corrected duration

Custom threshold: Tap the threshold value next to any profile to override it for your personal test results. Tap the refresh icon to restore the factory default.

14. Preferred Setups

Save named combinations of settings for instant recall with a single tap.

WHAT A SETUP STORES

- Name (e.g. "4×5 Ortho Sunny Day")
- Film stock + reciprocity profile
- Aperture
- Focal length
- Rolling Shutter Camera profile (optional)
- Custom Shutter Preset (optional)
- Toggle overrides (On / Off) for:
  - Environmental Compensation
  - RA4 Color Reversal
  - Positive Exposure
  - IR Compensation (when set to On, a wavelength sub-option appears: 720 nm / 850 nm / Don't set)

CREATING A SETUP

Go to **Settings → Preferred Setups** and tap **+**. Name the setup, choose which values to store, set any toggle overrides, and tap Save.

APPLYING A SETUP

Saved setups appear at the **top of the film stock picker** (above the full stock list). Tap a setup to apply all stored values at once.

EDITING & DELETING

In **Settings → Preferred Setups**, tap a setup to edit it or swipe left to delete it.

15. Positive Exposure

Adds **+0.5 stop** of exposure compensation for positive (reversal) processes such as tintypes.

- Selecting a tintype stock enables this automatically
- Can also be toggled manually via **Settings → Compensations → Positive Exposure**
- A sun icon chip appears on the meter screen when active
- Applies in addition to any other compensation

16. Bellows Compensation

For large format or macro photography using bellows extension, the meter automatically calculates the additional exposure needed.

Access: Tap the **Bellows button** in bottom bar row 2. The first tap enables bellows compensation and reveals an extension entry field below the control bar. Tap the field and enter the **extra extension beyond the focal length** in millimetres — this is the additional bellows draw beyond infinity focus, not the total bellows draw. When a non-zero value is set, a **+Xmm** chip appears in the status row confirming the compensation is active. Tap the Bellows button again to disable it and clear the value.

The compensation formula is:

> **Extra stops = 2 × $\log_2$((extra extension + focal length) ÷ focal length)**

Example: 150 mm lens with 30 mm of extra extension → 2 × $\log_2(180 \div 150) \approx +0.52$ stops. With 75 mm extra extension on the same lens → 2 × $\log_2(225 \div 150) \approx +1.17$ stops.

To clear: Tap the Bellows button again to disable compensation, or set the extension field back to 0.

17. Lens Filters

Access: Tap the **Filters button** in bottom bar row 2.

Select a filter to automatically apply its exposure compensation factor. Available filters include:

CATEGORY	FILTERS
Contrast	Yellow K2, Orange G, Red 25A, Green X1
Neutral density	ND2 through ND1000
Polarizer	Circular polarizer
Infrared	720 nm, 850 nm

The filter factor is automatically added to your exposure calculation.

IR Filters: The 720 nm and 850 nm slots are linked to the IR Light Estimator. Changing the wavelength in either place syncs the other automatically.

18. Spectral Sensitivity

Access: Settings → Compensations → **Spectral Sensitivity**

Adjusts the meter's readings to match the spectral response of your film.

MODE	FILM TYPE
Panchromatic	All modern films — responds to all visible light
Orthochromatic	Ilford Ortho Plus and similar — ignores red light
Blue-Sensitive	Early emulsions, cyanotype, Zebra dry plates — responds mainly to blue/UV

TWO CORRECTION PATHS

1. **Real-time RGB analysis (primary):** When the camera is delivering live frames, the meter analyses the actual colour of the light and computes a spectral correction. This is the most accurate path and is shown as a **Spectral** badge on the viewfinder.
2. **Solar-elevation color temperature (fallback):** Derived from GPS location and the sun's current elevation. Only applies when RGB analysis is not active — the two corrections are never stacked.

PINPOINT SPECTRAL SCAN

In spot metering mode, a **Scan Scene** button appears next to the Spectral badge. Tapping it captures the light colour at the exact spot selected and locks that correction in place (shown as a 🔒 badge). Useful when the main subject has different light than the wider scene. The lock persists until you tap ✕ next to the badge, switch film stocks, or change spectral mode.

19. Environmental Compensation

Access: Settings → Compensations → **Environmental Compensation**

Automatically adjusts for altitude and UV conditions using your GPS location to determine elevation and UV index.

- Particularly useful for high-altitude landscape photography
- Last known coordinates are cached locally and reused across restarts as a fallback
- **Manual UV override:** Tap the UV field and enter a value (0–11+) to override GPS-based UV data. Tap Clear to return to automatic readings.

COLOR TEMPERATURE COMPONENT

For blue-sensitive and orthochromatic film, the meter also accounts for solar elevation:

- Warm light (golden hour, low sun) contains less blue/UV energy — film needs more exposure
- The correction is negative (reduces metered EV, calling for longer shutter or wider aperture)
- At midday with high solar elevation the correction approaches zero
- Automatically suppressed when live RGB spectral analysis is active (the two corrections are not stacked)

20. RA4 Color Reversal

Access: Settings → RA4 Color Reversal, or the **RA4** toggle on the Meter screen.

When enabled, the meter analyses the scene color temperature and suggests **CMY filter corrections** for RA4 color printing. Select your paper stock for film-specific recommendations.

- Filter packs show Cyan, Magenta, and Yellow values needed for neutral color balance
- Color temperature estimation needs a wider scene sample — use **center-weighted or average metering** for reliable results (not spot)
- During flash mode, a separate RA4 analysis runs on the flash peak frame

21. IR Light Estimator

Access: Settings → **IR Light Estimator** section

Estimates the current infrared light strength — useful for planning IR photography with 720 nm or 850 nm filters.

The estimator combines GPS-based solar elevation, UV index, and live spectral camera data to produce an **IR strength index (0–10)** shown as a label (Low, Moderate, High, Very High) and a bar graph.

FILTER SELECTION

Tap **720 nm** or **850 nm** to choose your IR filter. This is synced with the Lens Filter sheet on the Meter screen — changing the wavelength in either place syncs the other. Enabling IR Estimation automatically adds the currently selected filter to the meter; disabling it removes it.

EV ADJUSTMENT

The **"+ N stops"** chip shows the estimated additional exposure needed for the selected filter under current conditions. Use this as a starting point.

INDICATORS

- **Solar Elevation** — sun's angle above the horizon; a lower angle means more atmospheric IR scattering
- **UV Index** — higher UV generally means stronger IR output

OFFLINE MODE

UV index and solar elevation are estimated from latitude, season, and time of day when no internet connection is available.

22. Two-Point Calibration

Calibrate the app against a trusted light meter for maximum accuracy. Each camera and lens combination (front, rear main, ultra-wide, telephoto) gets its own independent calibration profile.

Access: Settings → Metering → **Calibrate Light Meter**

THE PROCESS

The guided calibration measures both a dark and a bright reference scene using one of five supported light sources:

- Incandescent bulb
- LED panel
- Phone screen
- Overcast sky
- Window light

The process computes gain and offset corrections for that specific camera/lens combination.

CALIBRATION GATE

A full-screen overlay prevents metering until the active camera is calibrated. Tap **Calibrate Now** to run the process immediately.

FINE-TUNE VS REFERENCE METER

After base calibration, an optional step appears: point both your phone and a trusted reference meter (e.g. Sekonic) at the same scene, enter the reference EV reading, and the app saves an **EV offset** to close any remaining gap. Return to this step at any time via **Settings → Metering → Fine-tune with Reference Meter**.

SW BADGE

A small amber **SW** label appears next to the Measured EV whenever the app is estimating from software rather than raw camera EXIF (e.g. briefly during camera startup or lens switching). Accuracy is slightly lower during this period — wait a moment for it to disappear before taking a critical reading.

23. Field Journal

Access: The **Journal** tab at the top of the screen.

SAVING AN EXPOSURE

Tap **SAVE** on the Meter screen. A confirmation screen appears where you can:

- Review all captured exposure data
- Add notes
- Attach the viewfinder photo

Tap **Save Entry** to confirm.

JOURNAL LIST

Each entry shows compact modifier chips below the exposure values — coloured icons for active lens filters, manual exposure compensation, bellows extension, and environmental compensation — so you can see at a glance what was applied without opening the full detail view.

When a multi-point exposure is saved, a numbered chip appears in this row. Tap it to expand an inline panel directly on the list card showing the individual EV recorded at each point (P1, P2, P3 ...).

VIEWING AN ENTRY

Tap any entry to open the full detail view, which shows:

- All exposure settings (ISO, aperture, shutter, EV)
- Film stock and format
- All active compensations
- RA4 filter pack data
- Attached photo with metering indicator overlay

The photo overlay shows an amber indicator marking the metering mode and spot/center-weighted area that was active at save time.

EXPORTING

From the detail view:

- **Save to Gallery** — saves the composite card (photo + data overlay) to your photo library as a JPEG with exposure data embedded in EXIF
- **Share** — opens the system share sheet with the composite card

DELETING AN ENTRY

Swipe left on any journal entry and tap **Delete**.

24. Countdown Timer

Access: When exposure is locked and the calculated shutter time exceeds 2 seconds, a small **timer icon** appears next to the shutter readout in the control bar. Tap it to open the Countdown Timer.

Counts down from the currently displayed shutter speed — useful for long exposures where you need to time the shutter manually.

Workflow:

1. Lock your exposure
2. Tap the timer icon
3. A 3-second preparation countdown runs first — open your shutter during this window
4. The main countdown begins with a visual arc ring
5. When time is up, the app vibrates and beeps to signal the exposure is complete
6. Tap **Cancel** at any time to stop

Reciprocity toggle: When reciprocity failure correction is active, the timer shows a toggle between the corrected duration and the original base time. The corrected (reciprocity) time is selected by default — this is the duration your shutter should stay open.

25. Extra Tools

Access all Extra Tools via **Settings** → **Extra Tools tab** (wrench icon).

25.1 DEPTH OF FIELD CALCULATOR

Enter your film format, focal length, aperture, and subject distance to calculate:

- Near limit of sharp focus
- Far limit of sharp focus
- Total depth of field
- Hyperfocal distance

The circle of confusion (CoC) is automatically derived from the selected film format. Distance can be entered in metric (mm/m) or imperial (inches/feet). Results update instantly.

Hyperfocal tip: When focused at the hyperfocal distance, everything from half that distance to infinity is acceptably sharp.

25.2 ALTERNATIVE PROCESSES METER

Supported processes with calibrated nominal ISOs:

- **Classic Cyanotype** (ISO ~0.01) — Prussian-blue prints from ferric salts; the original blueprint process.
- **New Cyanotype** (ISO ~0.05) — faster, higher-contrast variant using ferric ammonium oxalate.
- **Salt Print** (ISO ~0.01) — earliest photographic paper; silver chloride gives warm sepia tones.
- **Platinum/Palladium** (ISO ~0.003) — archival; rich neutral-to-warm tones with wide tonal range.
- **Van Dyke Brown** (ISO ~0.008) — iron-silver contact process; warm brown tones.
- **Kallitype** (ISO ~0.008) — iron-silver process; toneable to sepia, neutral black, or blue-black.
- **Gum Bichromate** (ISO ~0.02) — pigment and gum arabic; painterly results, multi-layer capable.

Setup:

1. Select your process
2. Select your light source: **Sunlight** (uses live UV index from GPS) or indoor (UV lamp, LED, fluorescent)
3. The meter calculates an estimated print time based on current UV conditions
4. Use the **±3 stop fine-tune control** to dial in your personal process

The live UV indicator shows the current index and whether readings are live (GPS) or estimated. Falls back to an offline estimate from latitude and time of day when UV data is unavailable.

Glass UV Compensation

Two toggles below the fine-tune slider account for UV-absorbing glass layers in the printing stack. Standard soda-lime float glass (3–4 mm) transmits approximately 70% of UV at 365 nm, absorbing the rest — each glass layer in the optical path therefore costs roughly **+½ stop** of additional print time.

TOGGLE	OPTIONS	WHEN TO USE "GLASS"
Negative Substrate	Plastic / Film · Glass Plate	Your negative is on a glass plate (ambrotype, wet-plate interpositive, glass lantern slide). Leave on *Plastic / Film* for conventional film base.
Contact Frame Top Plate	None · Glass	Your contact printing frame has a glass pressure plate over the negative. Set to *None* if you print with the frame open or under a UV-transparent acrylic lid.

The meter multiplies the computed print time by `1 / 0.70` per enabled layer (≈ +0.51 stops each). Enabling both layers costs approximately **+1 stop** in total. The correction is applied automatically — no additional fine-tuning for glass is required.

25.3 NEGATIVE DENSITY READER

Found inside the Alternative Processes Meter. Measures the transmission density of a developed negative held over the device screen.

Two-step workflow:

1. **No negative** — tap **Sample Baseline** with nothing over the screen. Captures bare screen brightness as a reference.
2. **Place negative** — lay the negative emulsion-down over the lit crop guide, then tap **Sample Negative**. Calculates density as $D = \log_{10}(\text{baseline} \div \text{transmitted light})$.

D value guide:

D VALUE	INTERPRETATION
< 0.5	Thin — underexposed or underdeveloped
0.5–1.0	Normal — standard print exposure
1.0–1.5	Dense — overexposed or overdeveloped
> 1.5	Very dense — significantly extended print times needed

The **stops correction** value shows how many stops of additional (or reduced) exposure your print needs relative to normal density.

Re-sample the baseline whenever you change screen brightness or move environments.

25.4 WET PLATE TIMER

Estimates how long you have after pulling a collodion-coated plate from the sensitizing tank before the emulsion dries, then counts that window down.

Estimation factors:

- **Temperature** — warmer air evaporates solvent faster (rate doubles roughly every 11 °C above 20 °C)
- **Humidity** — drier air shortens the window sharply (90% RH gives ~3× longer window than 50% RH)
- **Elevation** — lower air pressure at altitude speeds evaporation slightly
- **Wind Exposure** — None / Light / Strong (Strong can cut the window to ~⅓ of calm-air value)

When **Environmental Compensation** is enabled and GPS data is available, temperature, humidity, and elevation are read automatically (shown with a teal **Live** badge). Otherwise, manual sliders appear.

Timer:

- **Amber** — under 30 seconds remaining
- **Red + pulsing** — under 10 seconds remaining
- Tap **Stop** to pause, **Reset** to recalculate (useful if conditions change)

25.5 STEP WEDGE TOOL

Generates a bracketed series of timed exposures centered on a base value — ideal for film test strips, contact print calibration, and alt-process exposure testing.

Mode: Choose **Camera** (shutter-speed exposures) or **Alt Process** (print times from the Alternative Processes tool).

Setup:

- **Steps:** 3, 5, 7, or 9 (always odd — center step is the base)
- **Stop increment per step:** ⅓, ½, ⅔, or 1 stop
- **Base time:** In Camera mode, enter as a decimal in seconds (e.g. 0.0125 for 1/80 s, 2 for 2 s). In Alt Process mode, enter in MM:SS.

Import shortcut: In Camera mode, lock the meter on the Meter tab and tap **Import** — the locked shutter value fills automatically. In Alt Process mode, compute a print time first and tap **Import** to fill the base time.

Running the sequence:

- Tap **Run Sequence** → **Start Sequence** to begin the first exposure immediately
- Each step counts down with an arc ring and vibrates + beeps when complete
- Tap **Start Next Step** when ready — the sequence never advances automatically
- Tap **Skip Step** to mark the current step done early, or **Cancel** to stop

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25.6 RECIPROCITY CALCULATOR

Applies the reciprocity failure correction formula to a measured exposure time to output the corrected time you should actually use.

Mode:

- **Film Profile** — select from your saved reciprocity profiles (same profiles used by the live meter)
- **Manual** — enter a custom p-factor and threshold directly

Calculation: Enter the measured exposure time in seconds. If the time is at or below the threshold, a note confirms no correction is needed. If it exceeds the threshold, the result shows measured time, corrected time, and extra time added. Results update instantly.

25.7 EQUIVALENT LENS CALCULATOR

Converts focal lengths between any film format and the 35mm full-frame reference, and displays the angle of view for both.

Mode:

- **Format → 35mm** — enter an actual focal length on your chosen format and see its 35mm equivalent
- **35mm → Format** — enter a 35mm focal length and find the actual focal length needed on your chosen format

Results: Shows focal length, diagonal angle of view, and horizontal × vertical angles for both sides. Results update instantly.

25.8 PINHOLE CALCULATOR

Characterises a pinhole camera by calculating effective f-stop, optimal pinhole diameter (Rayleigh criterion), and image circle coverage.

F/Stop Mode: Enter pinhole diameter (mm) and focal distance (mm/cm/inches). Results show:

- Effective f-stop
- Optimal pinhole diameter for maximum sharpness at that focal distance
- Image circle diameter at 90° AoV
- Which standard film formats fit within it

Coverage Mode: Enter pinhole diameter and select a film format. Shows three recommended focal distances:

- **Standard** (53° diagonal AoV)
- **Wide** (80° diagonal AoV)
- **Extra Wide** (110° diagonal AoV)

Each preset shows recommended focal distance, effective f-stop, optimal pinhole diameter, and optimal f-stop.

25.9 FLASH GUIDE NUMBER CALCULATOR

Uses the guide number relationship (GN = Distance × Aperture) to find either the correct flash distance or the required aperture.

Mode:

- **Find Distance** — calculates how far the subject can be from the flash at a given aperture
- **Find Aperture** — calculates the required aperture at a given distance

Inputs:

- Guide Number (GN) — your flash's rated GN in metres or feet
- GN Reference ISO — the ISO at which the GN is rated (typically 100)
- Shooting ISO — actual shooting ISO (higher ISO increases effective GN)
- Flash Power — Full (1/1) through 1/128 in one-stop fractions
- Aperture (Find Distance mode) or Distance (Find Aperture mode)

Results show the effective GN after ISO and power adjustments, along with the final distance or aperture.

25.10 ROLLING SHUTTER CALCULATOR

Helps find the correct tension and aperture setting for focal-plane shutter cameras (Graflex SLRs, Speed Graphic press cameras).

Built-in camera profiles:

- Speed Graphic
- Auto Graflex 1 & 2
- R.B. Series BD
- R.B. Super D

You can also add your own custom camera profiles.

Workflow:

1. Select your camera profile
2. Lock your exposure on the Meter tab — the target speed appears at the top of the calculator
3. The table highlights the nearest available tension/aperture combination in amber

Show on Meter: Enable to display the recommended tension and aperture slot directly on the main meter readout as a compact chip.

Badge colours:

- **Green** — exact (or very close) match
- **Amber** — approximation (+ = slightly faster than metered, − = slightly slower)
- **Grey** — closest match is more than ⅓ stop away

- **Bulb (amber)** — metered time is longer than the slowest speed in the table — use bulb mode

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25.11 CUSTOM SHUTTER SPEEDS

Save named presets of non-standard shutter speeds for mechanical cameras with vintage or non-standard click stops.

Creating a preset: Tap + **Add Preset**, give it a name, then enter shutter speeds separated by commas or spaces. Use fractions (1/125) or decimals (0.5, 2). Duplicates are removed and speeds are sorted automatically.

Show on Meter: Enable to display a badge on the main meter screen showing the nearest speed from your preset to the currently calculated shutter speed.

Badge colours:

- **Green** — exact or very close match
- **Amber** — nearest available speed is within ⅓ stop (+ = faster than metered, − = slower)
- **Grey** — nearest match is more than ⅓ stop away

A custom shutter preset can be saved as part of a **Preferred Setup** for quick recall.

25.12 ZEBRA EXPOSURE GUIDE

Two Zebra Dry Plates-specific calculators:

Zebra Exposure Chart Calculator: Select month, time of day, lighting condition, aperture, and subject type to estimate exposure time using historical Zebra ISO 2 chart data.

In-Camera Test Strip Creator: Generates five exposure strips in a linear progression from a base value. Choose **Shutter** or **F-Stop** mode and a stop increment (⅓, ½, or full stop). The center strip (highlighted amber) is the base. In Shutter mode, cumulative add-times are shown so you can expose each strip sequentially in the darkroom.

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26. Gear Management & Customization

Access: Settings → **Gear Management**

This section covers every way to customize the dials, lists, and values that appear throughout the app.

26.1 FILM STOCKS

Access: Settings → Gear Management → **Film Stocks**

The film stock panel lists your active stock library. Three ways to add stocks:

Built-in stocks: The app ships with a curated set of common panchromatic, orthochromatic, and specialty stocks. They appear in the stock picker automatically.

Zebra Dry Plates Film Stocks: Tap this button to browse the full Zebra Dry Plates catalogue fetched live from GitHub (cached for 24 hours — works offline after the first load). Tap + next to any stock to add it to your local library. Stocks include dry plates, tintypes, and golden tintypes, all at ISO 2 with blue-sensitive spectral profiles. Custom formats defined in the CSV (via formatWidth/formatHeight columns) are automatically registered as custom film formats when you add the stock.

Popular Film Stocks: Tap this button to browse a curated list of commonly used third-party stocks (Kodak, Ilford, Fuji, etc.) and add them with one tap.

Adding a fully custom stock: Tap + **Add Film Stock** to manually enter:

- Name
- ISO speed
- Spectral sensitivity (Panchromatic / Orthochromatic / Blue-Sensitive / Infrared / Cyanotype)
- Reciprocity profile (pick from existing profiles or leave none)
- Purchase URL (optional — shows a cart icon in the stock picker)
- Positive Exposure flag (auto-enables the +0.5 stop compensation when this stock is selected)

Removing a stock: Swipe left on any custom stock entry, or tap **Remove** inside the stock detail.

26.2 CUSTOM RECIPROCITY PROFILES

Access: Settings → Gear Management → Film Stocks → + **Add Film Profile**

Add a reciprocity failure profile for any film not already in the list.

Inputs:

- **Name** — identifies the profile in the picker
- **p-factor (Schwarzschild exponent)** — typical range 1.1–1.5. Higher values mean more severe failure (more extra time needed). Examples:
    - Kodak T-Max 100: ~1.26
    - Ilford HP5: ~1.31
    - Fuji Provia 100F: ~1.33
    - Kodak Portra 400: ~1.22
    - Wet plate collodion: ~1.3–1.5 depending on formulation
- **Threshold (seconds)** — the exposure time below which no correction is applied (default 1 s for most films; some slow emulsions start earlier at 0.5 s)

Using the profile: Once saved it appears in the reciprocity profile picker for any film stock. You can also use it directly in the Reciprocity Calculator (Extra Tools) without linking it to a stock.

26.3 FOCAL LENGTHS (LENS DIAL)

Access: Settings → Gear Management → **Lenses**

Controls which focal lengths appear in the focal length selector on the Meter screen.

Enabling / disabling built-in focal lengths: Tap any focal length chip to toggle it on (filled) or off (outline). The chip turns amber when active. Disabled focal lengths are hidden from the selector — use this to keep the list uncluttered if you only shoot a few lenses.

Built-in focal lengths available: 8mm, 10mm, 14mm, 15mm, 17mm, 18mm, 20mm, 21mm, 24mm, 28mm, 35mm, 40mm, 45mm, 50mm, 55mm, 65mm, 75mm, 80mm, 85mm, 90mm, 100mm, 105mm, 110mm, 120mm, 127mm, 135mm, 150mm, 180mm, 210mm, 240mm, 300mm, 360mm, 400mm, 450mm, 600mm, 800mm

Adding a custom focal length: Tap + **Custom Lens** and enter any focal length in mm (e.g. 58, 90, 165). Custom focal lengths appear in the lens dial alongside built-in values, sorted numerically. Use this for unusual LF lenses, vintage lenses, or any focal length not in the built-in list.

Removing a custom focal length: Tap the trash icon next to any custom focal length in the list. Built-in focal lengths cannot be permanently deleted — only toggled off.

How focal length affects the app:

- Drives the viewfinder frame line scale (field of view simulation)
- Used in bellows compensation formula (total draw ÷ focal length, where total draw = extra extension + focal length)
- Used in DoF calculations (hyperfocal distance, near/far limits)
- Determines which physical phone camera lens Auto Lens Selection picks

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26.4 FILM FORMATS

Access: Settings → Gear Management → **Formats**

Controls which film formats appear in the format selector on the Meter screen.

Enabling / disabling built-in formats: Tap any format chip to toggle it on or off. Enabled formats appear in the format picker; disabled ones are hidden. Keeping only the formats you shoot makes the picker faster to navigate.

Built-in formats available:

FORMAT	FRAME SIZE
35mm	24×36 mm
35mm Half Frame	18×24 mm
6×4.5	56×42 mm
6×6	56×56 mm
6×7	56×70 mm
6×9	56×84 mm
6×12	56×112 mm
6×17	56×170 mm
4×5 in	95×119 mm
5×7 in	127×178 mm
8×10 in	203×254 mm
11×14 in	279×356 mm
Instax Mini	46×62 mm
Instax Wide	62×99 mm
Instax Square	62×62 mm
110	13×17 mm
126	28×28 mm
127 (4×4)	40×40 mm
Disk	8×11 mm
APS-C	25×17 mm

Adding a custom format: Tap + **Custom Format**, enter:

- **Name** — what appears in the format picker (e.g. "9×12 cm")
- **Width** (mm) — short dimension of the frame
- **Height** (mm) — long dimension of the frame

The app derives the aspect ratio and viewfinder framing automatically. Custom formats also drive DoF calculations with a circle of confusion derived from the diagonal.

Removing a custom format: Tap the trash icon next to any custom format. Built-in formats cannot be permanently deleted — only toggled off.

How format affects the app:

- Sets the viewfinder frame line aspect ratio
- Sets the circle of confusion for DoF calculations
- Used in the Equivalent Lens Calculator (crop factor)
- Used in the Pinhole Calculator (image circle coverage)

26.5 APERTURES (F-STOP DIAL)

Access: Settings → Gear Management → **Apertures**

Controls which f-stops appear in the aperture selector on the Meter screen.

Enabling / disabling built-in f-stops: Tap any aperture chip to toggle it on or off. Disabled values are hidden from the selector. Use this to keep the list clean — for example, disable f/6.7 and f/9.5 if you never use non-standard LF lens apertures.

When you change Stop Increment (Settings → Metering → Stop Increment): new standard values for the newly selected increment are automatically merged into your active aperture set. For example, switching from full stops to ⅓ stops adds f/1.1, f/1.2, f/1.4 (⅓-stop steps between f/1 and f/1.4), etc.

Activating multiple stop increments simultaneously: Tap multiple stop increment options to activate more than one at once (e.g. both ⅓ stop and Full stop). The aperture and shutter speed selectors merge and deduplicate values from all active sets, sorted numerically. Snapping finds the nearest value across the combined list.

Adding a custom aperture: Tap + **Custom Aperture** and enter any f-number (e.g. 6.3, 9.5, 4.5). Useful for:

- Large format lenses with non-standard aperture markings (e.g. f/6.3, f/11.3)
- Older lenses with Continental stop series (f/3.5, f/4.5, f/6.3, f/9, f/12.5)
- Pinhole cameras where you've calculated an effective f-stop
- Any lens with a maximum aperture that falls between standard values (e.g. f/1.2 on a fast 50)

Removing a custom aperture: Tap the chip in the custom apertures row to remove it. Built-in apertures cannot be permanently deleted — only toggled off.

26.6 CUSTOM ISO VALUES

Access: Settings → Metering → **Custom ISO**

Add non-standard ISO values not on the built-in dial.

Adding a custom ISO: Tap + **Add ISO** and enter any value. Supports:

- Integers (e.g. 12, 400, 3200)
- Decimals (e.g. **0.3** for wet plate collodion, **5.3** for Zebra Dry Plates, **25.6** for some glass plates)
- Values below 1 are fully supported — useful for extremely slow processes

Custom values are merged with the standard ISO list, deduplicated, and sorted. They appear on the ISO dial alongside the built-in values.

Removing a custom ISO: Tap the chip in the custom ISO row to remove it.

ISO and film stock interaction: When you manually change the ISO dial after selecting a film stock, the stock is deselected and your manual ISO takes over. To return to stock-native ISO, reselect the film stock from the picker.

26.7 PREFERRED SETUPS

Create, edit, and delete named setting combinations — see Section 14 for full details.

26.8 VIEWFINDER CALIBRATION

Access: Settings → Viewfinder → **Viewfinder Calibration**

Sets the equivalent focal length of the phone camera being used. This is separate from the light meter calibration and only affects how the viewfinder frame lines are scaled. The default values are:

LENS	DEFAULT EQUIVALENT FL
Ultra-wide (iOS)	~13 mm
Main / Standard	~26 mm
Telephoto (iOS)	~77 mm
Android wide	~26 mm

Adjust these values if your specific phone's lens does not match the default — for example, some phones have a main camera at 24 mm equivalent rather than 26 mm.

To calibrate: Use the – / + buttons to step the value in 1 mm increments, or tap the field to type a value directly.

27. App Tools Configuration

Access: Settings → Preferences → **App Tools Configuration**

Show or hide optional features to keep the interface focused on what you use.

SETTINGS FEATURES (SHOWN/HIDDEN FROM THE SETTINGS TAB)

FEATURE	WHAT IT CONTROLS
Spectral Sensitivity	Film-type spectral corrections
Positive Exposure	+½ stop for reversal processes
Environmental Compensation	UV/altitude corrections
RA4 Color Reversal	Filter pack suggestions
IR Light Estimator	Infrared exposure guidance
Viewfinder Zoom Slider	Left-edge focal length slider
UV Lens Coating	Scale UV compensation by lens coating type *(requires Fight Club unlock — see Section 31)*

EXTRA TOOLS (SHOWN/HIDDEN FROM THE EXTRA TOOLS TAB)

TOOL	NOTES
Zebra Exposure Guide	Always visible
Depth of Field Calculator	Always visible
Alternative Processes Meter	Always visible
Wet Plate Timer	Always visible
Step Wedge Tool	Always visible
Rolling Shutter Calculator	Always visible
Reciprocity Calculator	Always visible
Equivalent Lens Calculator	Always visible
Pinhole Calculator	Always visible
Flash GN Calculator	Always visible
Parallax Calculator	*(requires Fight Club unlock — see Section 31)*
LF Movements	*(requires Fight Club unlock — see Section 31)*
Zone System	*(requires Fight Club unlock — see Section 31)*
Scene DR Indicator	*(requires Fight Club unlock — see Section 31)*
Custom Shutter Speeds	Always visible

When all Extra Tools are hidden, the Extra Tools tab shows an empty state prompt.

All toggles default to on. Changes take effect instantly and persist across app restarts.

28. Settings Reference

COMPENSATIONS SECTION

SETTING	DESCRIPTION
Spectral Sensitivity	Panchromatic / Orthochromatic / Blue-Sensitive
Positive Exposure	+0.5 stop for reversal processes
Environmental Compensation	Auto UV/altitude from GPS
Bellows Compensation	Macro/LF extension penalty
Manual Exposure Compensation	±EV offset applied to all readings

METERING SECTION

SETTING	DESCRIPTION
Stop Increment	⅓ stop, ½ stop, ⅔ stop, full stop (can activate multiple simultaneously)
Spot Metering Size	Controls the radius of the spot circle (small = pinpoint, large = wider spot)
Center-Weighted Size	How much of the centre is prioritised vs. the edges
Average Metering Size	100% = full frame; lower = centre-biased average
Multi-Point Metering	Enable averaged 2–4 point readings
Custom ISO	Add non-standard ISO values including decimals
Flash Trigger	Off / Optical Slave (LED) / Cable Sync (3.5mm) / Light Sensor (Android)
Torch Duration	LED burst length for Optical Slave mode: 50 ms / 100 ms / 150 ms (default) / 200 ms
Flash Sensitivity	Maximum / High / Normal (default) / Reduced / Low / Minimum — see Section 11
Metering Quality	Speed vs. Balanced pipeline — affects snapshot rate and analysis depth
Calibrate Light Meter	Two-point calibration wizard (per camera/lens combination)

VIEWFINDER SECTION

SETTING	DESCRIPTION
Camera Lens	Ultra-wide / Main / Telephoto (iOS)
Auto-select Camera Lens	Automatic lens switching by focal length
Zoom Slider	Enable/disable focal length slider
Viewfinder Calibration	Per-lens viewfinder calibration
Viewfinder Margin	Inner frame line margin (0–20%)
Orientation	Portrait / Landscape

PREFERENCES SECTION

SETTING	DESCRIPTION
Units	Metric / Imperial
Language	English / Slovenian / Spanish / French / German / Japanese / Chinese Simplified
Keep Screen Awake	Prevents sleep during metering
Show Color Temperature	Live Kelvin badge on viewfinder
Debug Logging	Detailed internal logs for bug reports
App Tools Configuration	Show/hide optional features
Theme	Dark / Light

ABOUT SECTION

- App version and build number
- Website link
- Report a Bug (opens native email compose)
- Debug Log (view / copy / clear internal logs)
- Factory Reset — erases ALL data (saved exposures, calibration, custom stocks, settings). **This cannot be undone.**

29. Settings Transfer

Access: Settings → **Export / Import Settings**

Export your complete settings (calibration profiles, custom stocks, preferred setups, film formats, custom lenses, apertures, and all preferences) to a JSON file. Import this file on another device to replicate your setup exactly.

Export: Tap **Export Settings** — a share sheet opens with the settings file. Save it to Files, AirDrop it, or email it.

Import: Tap **Import Settings** — select a previously exported settings file. The app validates the file and applies all settings. Existing settings are replaced.

> **Tip:** *Use Export / Import Settings to back up your calibration profiles before reinstalling the app or switching devices.*

30. Troubleshooting

BLANK OR DARK VIEWFINDER

- Ensure camera permission is granted in your device's Settings → Privacy → Camera
- Try switching camera lenses in Settings → Viewfinder → Camera Lens
- Restart the app if the camera feed is frozen unexpectedly

SW BADGE APPEARING CONSTANTLY

The **SW** badge means the app is using software EV estimation rather than raw camera EXIF. This is normal briefly during startup or lens switching. If it persists, try re-calibrating via Settings → Metering → Calibrate Light Meter.

EXPORT FAILS SILENTLY

- Ensure media library permission is granted (Settings → Privacy → Photos)
- If the image hasn't fully loaded before export, the app will wait and display a message
- If the image file was deleted or moved since saving, the app will show a specific "file not found" error

CALIBRATION GATE BLOCKING THE VIEWFINDER

The calibration gate appears when the active camera has not been calibrated. Tap **Calibrate Now** to run the two-point calibration wizard. After calibrating each lens, the gate will not reappear for that lens.

FLASH NOT DETECTED

- For Optical Slave mode, point the phone toward the flash unit's optical sensor window
- Lower **Flash Sensitivity** (Settings → Metering → Flash Sensitivity) toward Maximum or High if the flash isn't registering
- Raise it toward Reduced or Low if you are getting false triggers from ambient light changes
- After changing sensitivity, **re-arm the flash** for the new setting to take effect

RECIPROCITY CORRECTION NOT SHOWING

The corrected time only appears after you **Lock** the exposure. During live metering, the raw calculated time is always shown. Lock first to see the correction.

JOURNAL PHOTOS NOT EXPORTING CORRECTLY (IOS)

After an iOS app update, the image file path may have changed. The app automatically normalises stored file paths on startup — if an entry still shows a blank photo, delete and re-save that reading.

31. Fight Club

Fight Club is a hidden tier of advanced tools built into the app. The features here are intentionally separated from the main manual so this section can be shared selectively with people who have access.

HOW TO UNLOCK

1. Go to the **Settings** tab
2. Tap the **SHP splash logo image** (the photo at the top of the About section) **9 times**
3. Each tap must happen within **2 seconds** of the previous one — if you pause for more than 2 seconds the counter resets and you must start again
4. On the 9th tap, a confirmation modal appears and Fight Club is unlocked
5. All Fight Club features appear immediately — no restart required

To lock Fight Club again: repeat the same 9-tap sequence. A second modal confirms the lock.

WHAT FIGHT CLUB UNLOCKS

Fight Club reveals five features that are completely hidden from the standard interface:

FEATURE	WHERE IT APPEARS
UV Lens Coating	Settings → Compensations / App Tools Config
Zone System	Settings → Metering toggle + Meter screen bottom bar
Scene DR Indicator	Settings → Metering toggle + histogram
LF Movements	Settings → Extra Tools + Meter screen bottom bar
Parallax Calculator	Settings → Extra Tools

Each feature can be individually toggled on or off via **Settings → App Tools Configuration** once Fight Club is unlocked. If you toggle a feature off, it disappears from the interface — but Fight Club itself remains unlocked.

UV LENS COATING

Access: Settings → Compensations → **Lens UV Coating** *(visible only when Orthochromatic or Blue-Sensitive spectral mode is active)*

Modern multi-coated lenses block most UV-A, which directly reduces the UV compensation that Environmental Compensation would otherwise apply. This tool lets you scale that compensation to match your actual lens.

Toggle the switch to enable the correction, then tap **Coating Type** to choose:

COATING TYPE	UV-A TRANSMISSION	TYPICAL LENSES
UV-transparent	~92%	Quartz optics, scientific lenses
Uncoated	~85%	Reference point — no correction applied
Single-coated	~65%	1950s–60s lenses (Leica M pre-1970, Nikkor-S)
Multi-coated / MC	~35%	Most 1970s–90s SLR lenses
Super multi-coated / Nano	~12%	Modern Pentax SMC, Nikon Nano Crystal, Canon ASC
UV-cut	~3%	Portrait telephotos, fluorescent-corrected optics
Custom %	—	Enter a known UV-A transmission value directly

The UV component of the environmental correction is scaled by the selected transmission fraction and shown as a separate **"Lens coating: –X.X EV"** line in the Compensations readout.

ZONE SYSTEM

Access: Tap the **Zone System button** in bottom bar row 2 (contrast icon), or toggle via Settings → Metering → **Zone System**

Applies Ansel Adams-style zone placement to the meter — you place a specific tone at a specific zone and the meter adjusts the exposure accordingly.

Workflow:

1. Tap the Zone System button on the Meter screen
2. A bottom sheet slides up showing the zone strip and options
3. Tap your **Placement Mode** — **Single zone** (place any tone at Zone V to Zone VIII) or **Shadow/Highlight** (lock both ends independently)
4. Point at the area you want to place and tap **Set** — the meter reads that zone and calculates the EV adjustment needed
5. The Zone System button shows the current adjustment (e.g. **+2** or **−1**) so you always know it's active
6. The meter automatically switches to **Spot mode** while Zone System is active — spot metering gives the most precise tonal placement
7. Tap **Clear** in the sheet to remove the placement and return to normal metering

The EV adjustment from zone placement is added to your exposure in the same way as manual compensation.

SCENE DR INDICATOR

Access: Settings → Metering → **Scene DR Indicator** (toggle), then visible below the histogram on the Meter screen

Shows a live bar comparing the dynamic range of the scene in front of you against the DR capacity of your selected film stock.

Reading the bar:

- **Green** — scene fits well within the film's range
- **Amber** — scene is approaching the film's limit (above 85% of capacity)
- **Red** — scene DR exceeds what the film can hold — expect blocked shadows or blown highlights

Stop readout: "X.X / Y stops" — for example, "9.2 / 11 stops" means the scene spans 9.2 stops and your film holds 11, leaving 1.8 stops of headroom.

Tap the bar to open a brief explanation of the numbers.

LF MOVEMENTS

Access: Tap the **LF Movements button** in bottom bar row 2 (compare icon), or toggle via Settings → Extra Tools → **LF Movements**

Opens a live overlay on the viewfinder for planning and documenting large-format camera movements. The meter locks automatically when you open the overlay so the reading stays frozen while you work through the movements.

Front standard (lens board):

- **Rise / Fall** — moves lens up or down (keeps verticals parallel for architecture)
- **Shift** — moves lens left or right (panoramic stitching, avoiding obstructions)
- **Tilt** — rotates the lens about a horizontal axis (Scheimpflug plane of focus control)
- **Swing** — rotates the lens about a vertical axis

Rear standard (film back): Same movements, but rear movements change both focus and perspective. Rise/fall on the rear standard introduces convergence or divergence of verticals.

Scheimpflug Principle: The tool illustrates the relationship between film plane, lens plane, and subject plane, and calculates the tilt angle required for a given subject plane angle.

Bellows Focus Calculator: Also found inside the LF Movements panel. Given a subject distance and focal length, it calculates:

- Total bellows draw needed to focus at that distance
- Extra extension beyond the focal length (the additional draw above infinity focus)
- Resulting exposure penalty in stops

Tap **Apply to Bellows Compensation** to send the calculated **extra extension** value directly to the bellows field on the Meter screen.

Tap the LF Movements button again (or tap outside the overlay) to close the panel. The Lock state returns to what it was before you opened it.

PARALLAX CALCULATOR

Access: Settings → Extra Tools → **Parallax Calculator** (scroll to the bottom of the Extra Tools tab)

Calculates the framing correction needed to account for the offset between your viewfinder window and the taking lens. This parallax error is most noticeable at close subject distances.

Lens offset by camera type:

CAMERA TYPE	CORRECTION NEEDED
Rangefinder (Leica, Voigtländer, Canonet)	Yes — measure VF window to lens centre
TLR (Rolleiflex, Mamiya C)	Yes — Y offset = lens centre-to-centre spacing (~45–48 mm)
Press camera with optical VF	Yes — measure VF window to lens centre
SLR	No — you view through the taking lens
Large format view camera	No — you compose on the ground glass

Horizontal offset (X): positive = lens is to the right of the viewfinder **Vertical offset (Y):** positive = taking lens is below the viewfinder

Enter your lens-to-VF offsets and subject distance. The calculator outputs the frame shift in mm at the film plane, and the corrected aim point so your viewfinder composition matches what the lens actually records.

ProFilm Meter is developed in collaboration between Scott Henderson Photography and Zebra Dry Plates. All metering and analysis runs entirely on-device — no data is collected, transmitted, or stored on external servers.

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